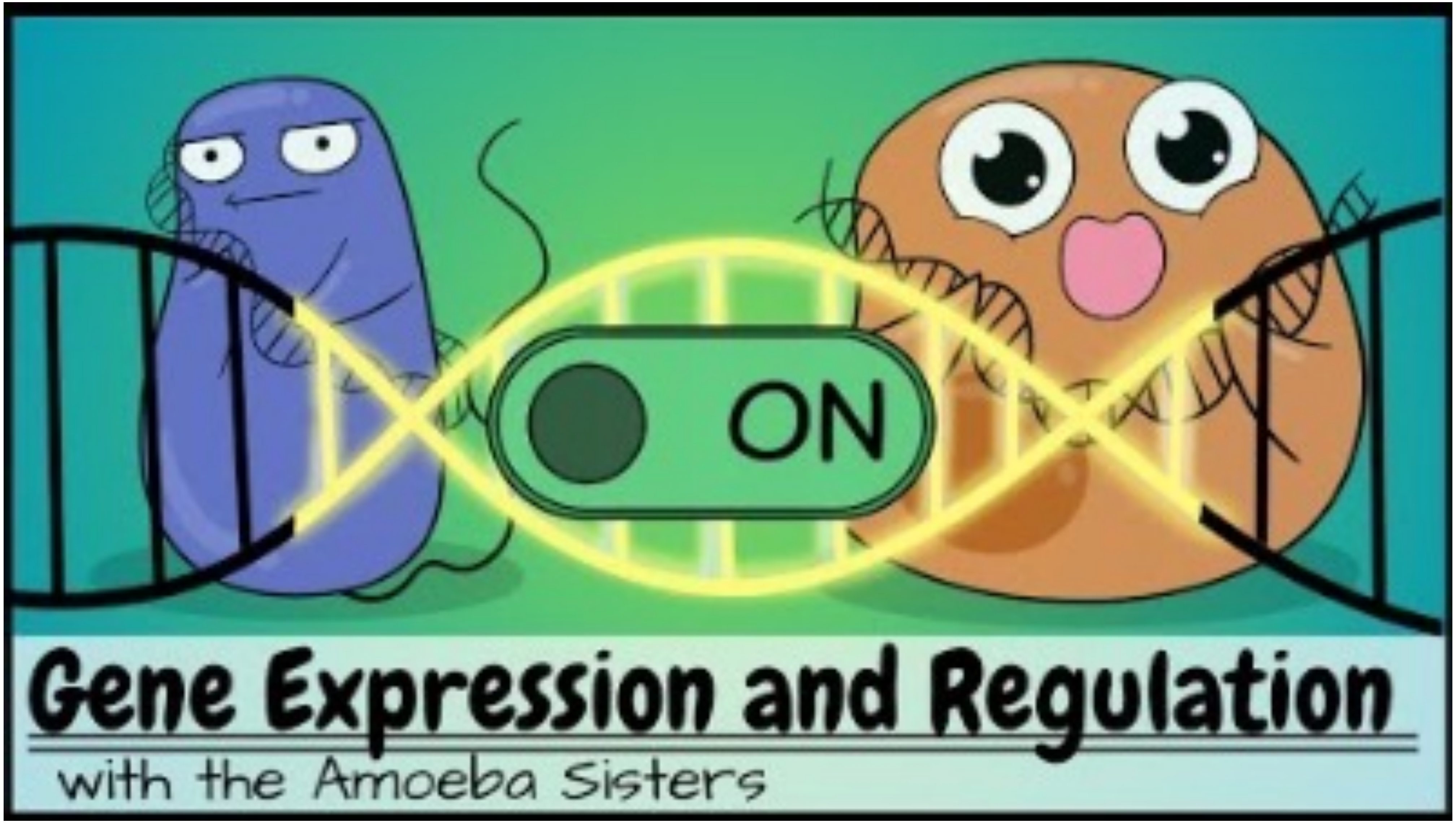


https://www.youtube.com/watch?v=ebIpkw3XapE

Control in living systems

Gene expression

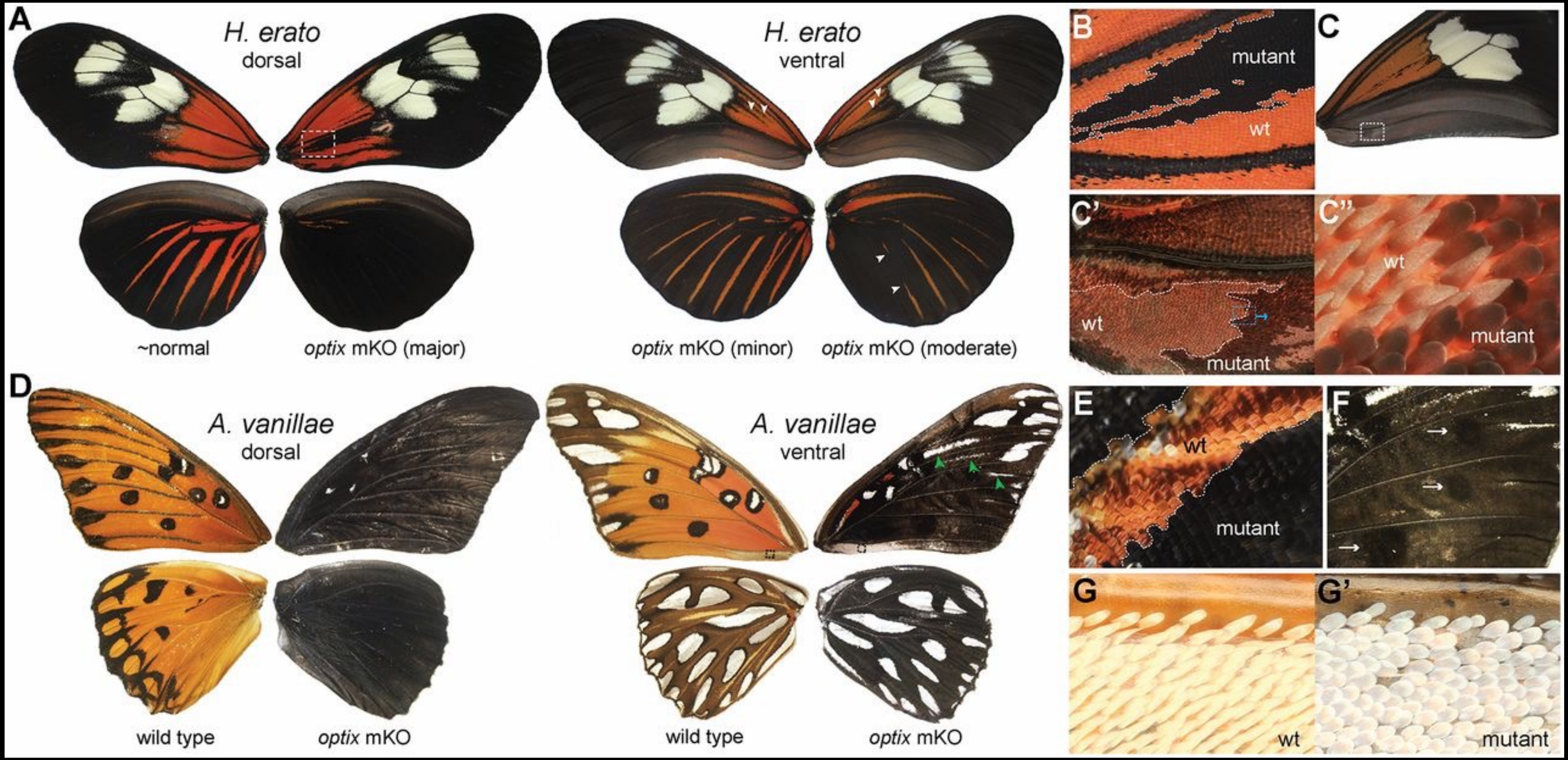
control of cell division

Hormones, signals & receptors

feedback control

control failure & disease

What do you see here??



https://doi.org/10.1073/pnas.1709058114

Gene expression

Information encoded in a gene is turned into a function.

on/off switch” to control when and where RNA molecules and proteins are made and as a “volume control” to determine how much of those products are made.

carefully regulated, changing substantially under different conditions and cell types.

The RNA and protein products of many genes serve to regulate the expression of other genes.

Where, when, and how much a gene is expressed can also assessed by measuring the functional activity of a gene product or observing a phenotype associated with a gene.

The DNA strand that corresponds to the mRNA is called the coding or sense strand.

Gene expression

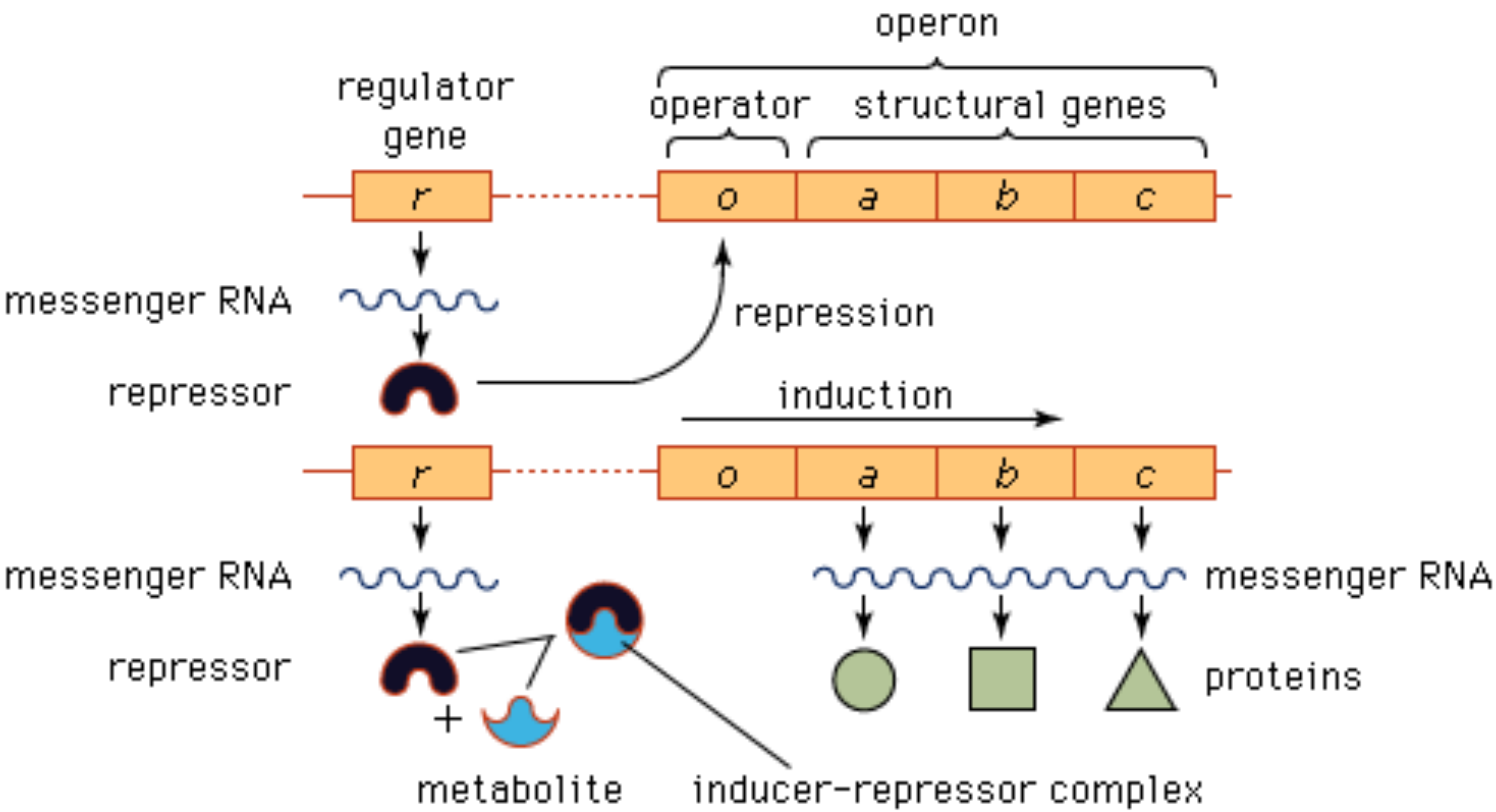
Operon – grouped genes that are transcribed together – code for functionally similar proteins

Promoter – section of DNA where RNA polymerase binds
Operator – Controls activation of transcription
- on/off switch
- between promoter and genes for proteins – structural genes

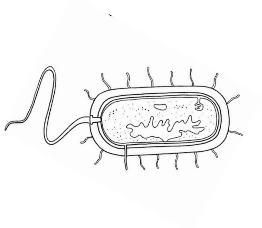
Repressor protein – binds to operator to block RNA polymerase and shut down transcription
- Turns off the operon

Corepressor – keeps the repressor protein on the operator (Trp operon)
Inducer – pulls repressor off the operator
-Turns on the operon – lactose on the lac operon

Regulatory gene – produces the repressor protein
Structural genes – code for proteins


operon
regulator
gene
operator
structural genes
r
o
a
b
c
messenger RNA
repression
repressor
induction
r
o
a
b
c
messenger RNA
messenger RNA
repressor
proteins
+
metabolite
inducer-repressor complex

Lac Operon



No lactose

Expression of lactose metabolism genes X

Gene z: Galactosidase

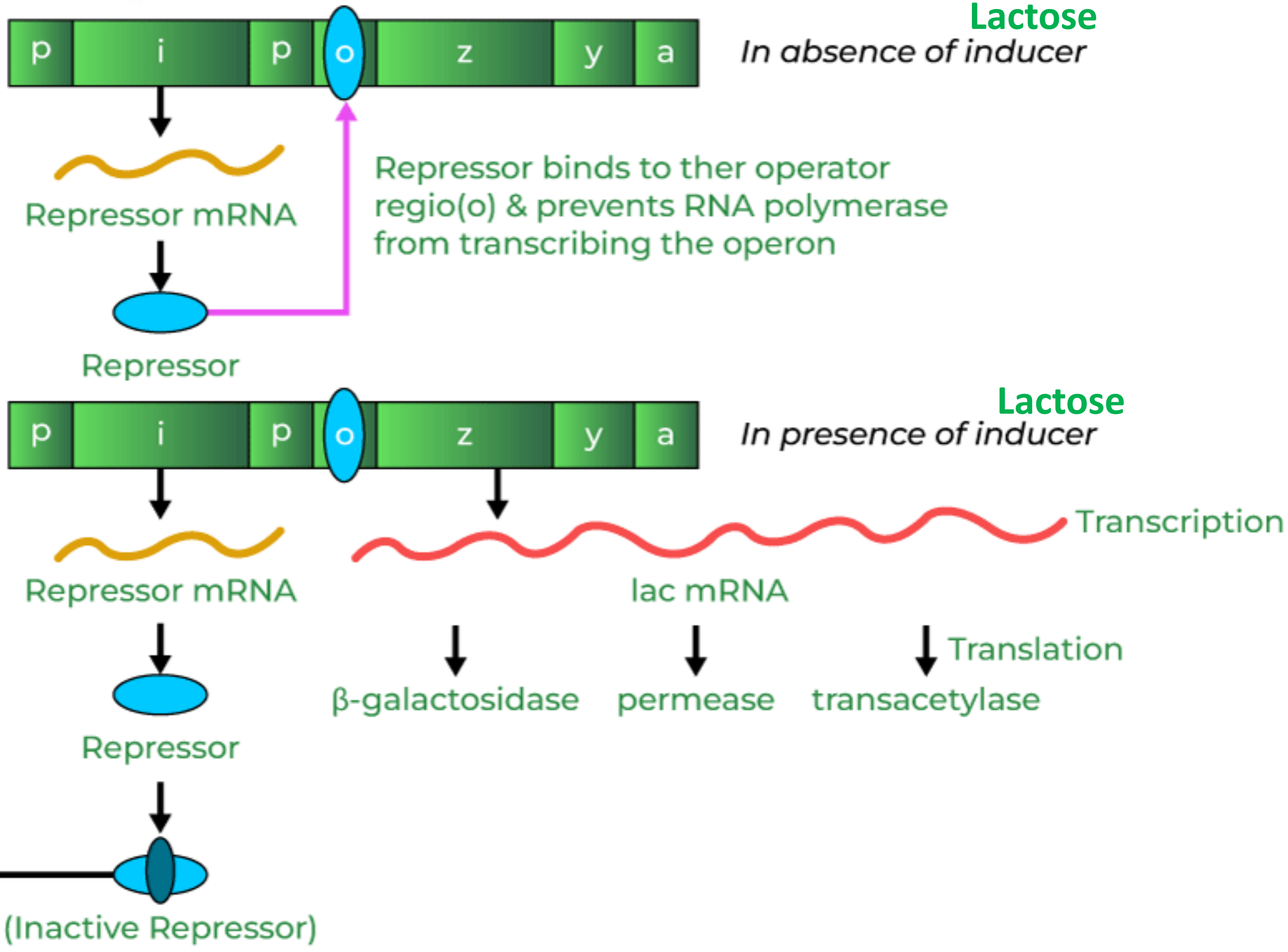
Lactose- Glucose

Gene y: Permease

Lactose crossing membrane

Gene a: Transacetylase

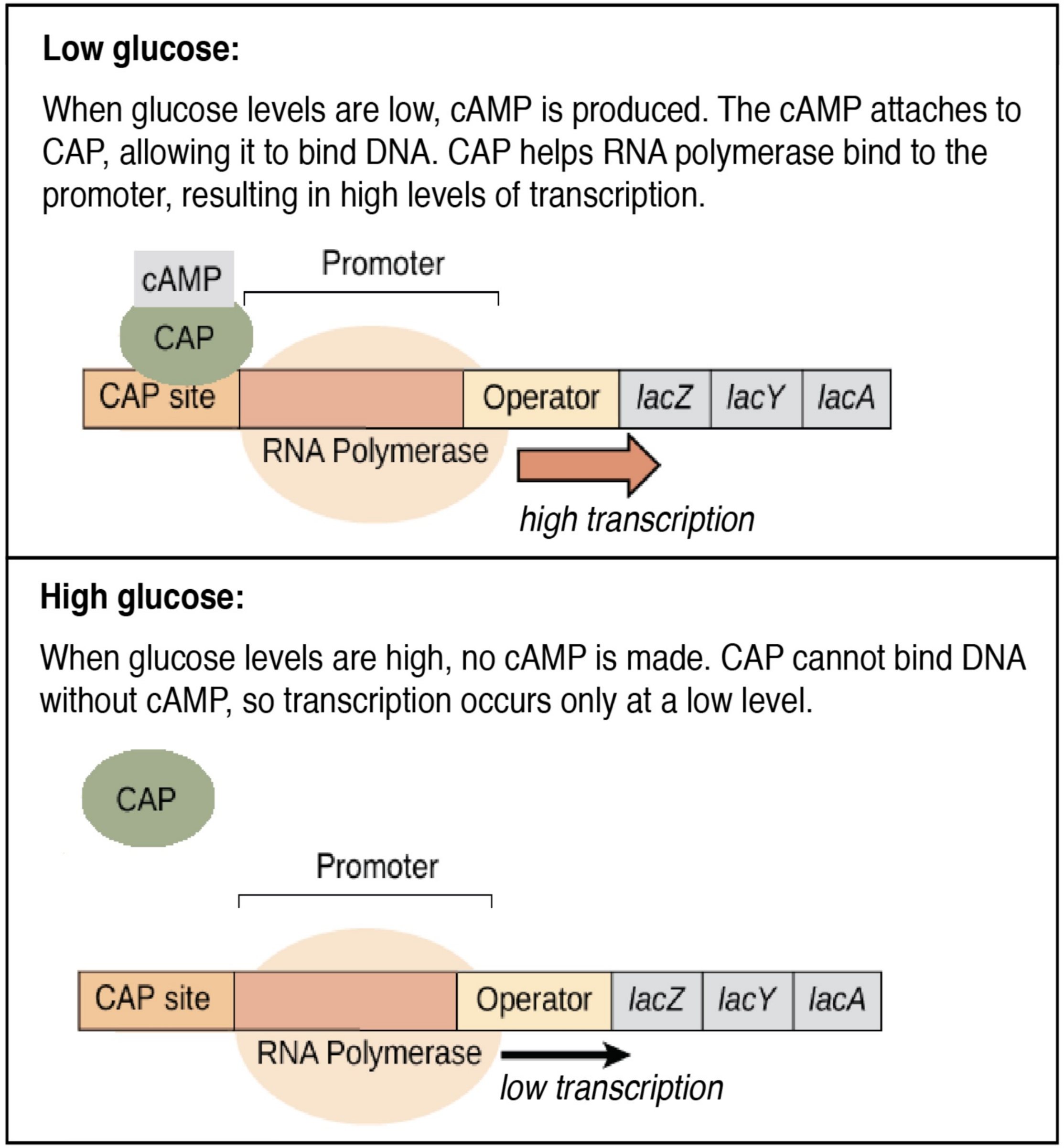
Helps Galactosidase



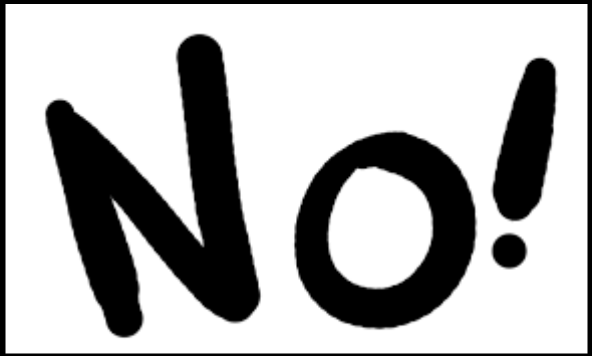
Lactose

Does the gene expression occur at a constant rate?

- **Catabolite activator protein** (**CAP**) is a positive regulator
- Low glucose (Hunger Signal) cAMP enables CAP and DNA binding
- CAP increases the rate of gene expression



Does all the DNA gets transcripted?



Only certain sections of the DNA are made (transcribed) into message (mRNA)

AND…only part of the mRNA is actually used and sent out of the nucleus to meet up with a ribosome!

This is EDITING!!

How is mRNA Edited?

On a mRNA strand there are areas called: Exons and Introns

Introns are cut out before leaving the nucleus

Exons are left, and this shortened piece of mRNA leaves the nucleus and gets Translated into Proteins

RNA Processing Regulation

Alternative RNA Splicing – different regions of the pre-mRNA serve as introns or exons creating different mRNA strands depending on what is removed & spliced together